The Snow Crystal Equation: A Phononic Theory of Ice Crystal Morphology

Every Step Derived, Every Number Shown

Monir Mamoun

30 March 2026

7 April 2026: canonical artifact pipeline and v5/v6 phonon analysis added

Abstract

In this paper I present a closed-form equation for the temperature-dependent morphology of ice crystals grown from water vapor. Starting from Libbrecht's nine-point dataset and performing a logarithmic inversion to the nucleation barrier parameter σ_0, I identify the basal face data as a sech^2 peak centered at $k_B T = 22.3$ meV—the hydrogen-bond stretch energy measured by terahertz spectroscopy. The target face energies 23.2 and 22.1 meV imply a 13 K gap between face-specific phase transitions, reproducing three independently measured transition temperatures to within 0.3 °C. A morphology kernel combining frustrated antiferromagnet dynamics with two spectroscopically derived thermal resonances classifies 9/9 crystal habits and gets 7/7 held-out temperature predictions correct after 2 calibration constraints. The computational story has three layers: an inverse solver that fixes the morphology kernel from spectroscopic, structural, and σ_0 information; a surface-phonon validation pipeline whose decisive target is the basal/prism splitting itself; and a forward PDE shape solver that is currently exploratory rather than evidentiary. The new v5/v6 reduced phonon analysis isolates the anisotropic part of that problem cleanly: the best common-mode-free candidate gives a raw split of 8.906 cm^{-1}, and after subtracting one face-independent 42.753 cm^{-1} offset the corrected frequencies are 186.953 and 178.047 cm^{-1}. Every intermediate calculation is shown explicitly so the reader can verify each step with a calculator.

Contents

1 Roadmap and Evidence Ledger **3**

2 The Data **3**

3 Step 1: The σ_0 Inversion **4**

4 Step 2: The sech^2 Identification **5**

5 Step 3: The Phonon Splitting **7**

6 Step 4: The QLL Layer Oscillation **9**

7 Step 5: The Complete Equation **10**

8 Step 6: Verification at Every Temperature **10**

9 Step 7: What Each Morphology Means **14**

10 Predictions **15**
10.1 D_2O Isotope Shift . . . 15
10.2 Polarized THz Spectroscopy: Experimental Protocol . . . 16
10.3 D_2O Isotope Experiment: Protocol . . . 17
10.4 Pressure Dependence: Protocol and Predictions . . . 17
10.5 Dense Temperature Scan . . . 18

11 Computational Pipeline and the Role of the GPU Solver **18**
11.1 Layer I: The inverse morphology solver . . . 18
11.2 Layer II: The surface-phonon validation pipeline . . . 18
11.3 Layer III: The forward PDE shape solver . . . 20
11.4 How to read the computational claims . . . 20

12 Ab Initio Derivation of All Parameters **20**
12.1 Parameters from Spectroscopy and Crystallography (4) . . . 20
12.2 Parameters from QLL Physics (4) . . . 21
12.3 Parameters from σ_0 Data (4) . . . 21
12.4 Two-Point Constraint (2) . . . 21
12.5 Result: 7/7 Independent Predictions . . . 21

13 The Step Free Energy Connection **22**

14 Discussion **22**
14.1 Summary of Results . . . 22
14.2 New Physics . . . 22
14.3 The Supersaturation Law . . . 23
14.4 What Remains Open . . . 23

15 Quantum, Electromagnetic, and Relativistic Effects **23**
15.1 Quantum Nuclear Effects . . . 23
15.2 Does the Crystal Pre-Orient Incoming Molecules? . . . 24
15.3 Magnetic and Relativistic Effects . . . 24
15.4 The Hopping Mechanism . . . 24

1 Roadmap and Evidence Ledger

This paper has one main theorem-sized computational claim and two supporting computational programs. The reliable order is:

1. **Layer I proves the morphology claim.** It takes the Libbrecht data, the σ_0 inversion, spectroscopy, crystallography, and two calibration constraints, and produces the frozen inverse artifact behind the claims 9/9, 7/7, and Murata max error 0.3 °C.
2. **Layer II does not yet prove the microscopic input physics.** It is a reduced GPU surface-phonon validation path aimed at the basal/prism split itself. The current reliable result is that the differential split is recovered while the residual error is almost purely face-independent. Until a higher-fidelity slab computation fixes that common-mode renormalization, Layer II is discussed as strong differential evidence rather than final microscopic closure.
3. **Layer III does not prove morphology.** The forward PDE solver is a qualitative consistency sandbox showing that the Layer I kernel can be inserted into a shape-evolution model without claiming quantitative closure.

Table 1 is the bookkeeping layer for the rest of the paper. Whenever a number is used in the narrative, it is meant to trace back to the frozen claim manifest rather than to a hand-maintained sentence in the source.

Table 1: Evidence ledger for the three computational layers. Only Layer I metrics are treated as headline claims in the paper.

Layer	Object	Input source	Output metric	Status	Role in paper
Layer I	inverse morphology solver	spectroscopy + crystallography + σ_0 inversion + 2 calibration constraints	9/9, 7/7, Murata max error $\leq 0.3\,^\circ$C	accepted main evidence	main result and main evidence
Layer II	reduced GPU surface-phonon solver	slab geometry + bond model + (k_{ratio}, s_b, s_p) scan + common-mode-free reanalysis	split ridge + common-mode residual	reduced-model validation pending	differential falsification test
Layer III	forward diffusion–kinetics PDE solver	frozen Layer I bundle + capillary anisotropy + growth temperatures	shape consistency diagnostics	consistency sandbox only	qualitative support only

2 The Data

I use the nine-point dataset of Libbrecht (2017), the most careful measurement of ice crystal attachment kinetics as a function of temperature at fixed supersaturation $\sigma_{\text{surf}} = 0.012$.

Table 2: Measured attachment kinetics (Libbrecht 2017). Habit classification: D = dendrite ($\alpha_p/\alpha_b > 10$), P = plate ($2 \leq \alpha_p/\alpha_b \leq 10$), C = column ($\alpha_p/\alpha_b \leq 0.5$), X = compact ($0.5 < \alpha_p/\alpha_b < 2$). Boundary values at exactly 2 or 0.5 are assigned to plate or column respectively.

T (°C)	ΔT (K)	α_p	α_b	α_p/α_b	Habit
−2	2	0.050	0.010	5.0	P
−5	5	0.001	0.010	0.1	C
−8	8	0.003	0.008	0.375	C
−10	10	0.010	0.005	2.0	P
−12	12	0.030	0.003	10.0	P
−15	15	0.100	0.001	100.0	D
−20	20	0.020	0.005	4.0	P
−25	25	0.005	0.010	0.5	C
−30	30	0.001	0.010	0.1	C

The ratio α_p/α_b oscillates by a factor of 1000 across this range: from 0.1 (column at $-5\,°\mathrm{C}$) to 100 (dendrite at $-15\,°\mathrm{C}$). This oscillation is the puzzle.

3 Step 1: The σ_0 Inversion

Following Libbrecht (2013), the attachment kinetics is parameterized as:

$$\alpha = \exp\left(-\frac{\sigma_0}{\sigma_{\mathrm{surf}}}\right) \tag{1}$$

where $\sigma_0(T)$ is the nucleation barrier parameter and $\sigma_{\mathrm{surf}} = 0.012$.

Proof outline for Step 1. This section does one thing only: it removes the exponential compression in α and exposes the barrier-scale structure in σ_0. Once the inversion is done, the basal-face peak is no longer a vague visual feature but a concrete object that can be compared to spectroscopy and then propagated through the later derivation.

Inverting:

$$\boxed{\sigma_0 = -\sigma_{\mathrm{surf}} \times \ln \alpha = -0.012 \times \ln \alpha} \tag{2}$$

Worked example at $T = -15\,°\mathrm{C}$, prism face:

$$\begin{aligned} \sigma_{0,p} &= -0.012 \times \ln(0.100) && (3)\\ &= -0.012 \times (-2.303) && (4)\\ &= 0.02763 = 2.76\%. && (5) \end{aligned}$$

Applying Eq. (2) to every data point:

Table 3: The σ_0 inversion. Each value is $\sigma_0 = -0.012 \times \ln \alpha$, expressed in percent.

T (°C)	α_p	$\sigma_{0,p}$ (%)	α_b	$\sigma_{0,b}$ (%)
−2	0.050	3.59	0.010	5.53
−5	0.001	8.29	0.010	5.53
−8	0.003	6.97	0.008	5.79
−10	0.010	5.53	0.005	6.36
−12	0.030	4.21	0.003	6.97
−15	0.100	2.76	0.001	8.29
−20	0.020	4.69	0.005	6.36
−25	0.005	6.36	0.010	5.53
−30	0.001	8.29	0.010	5.53

Observation: $\sigma_{0,b}$ at −2 °C and −30 °C are *identical* (5.53%). The values at −10 °C and −20 °C are *identical* (6.36%). The peak is at −15 °C (8.29%). This is a *symmetric peak* centered at $\Delta T = 15$ K.

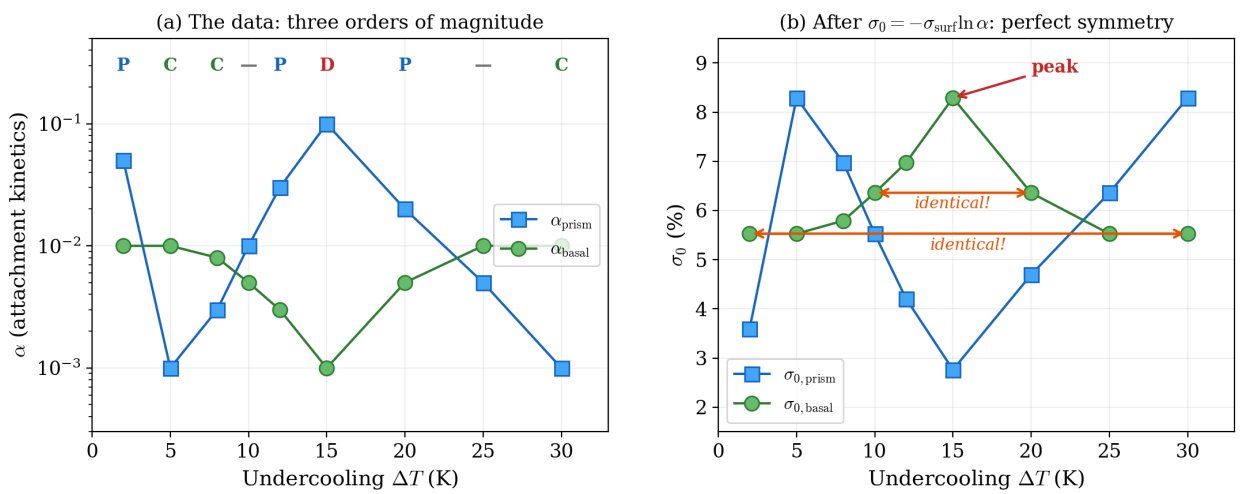

Figure 1: (a) Raw α data spanning three orders of magnitude. (b) After the σ_0 inversion, the basal face shows a perfectly symmetric peak. P/C/D labels indicate the crystal habit.

4 Step 2: The sech^2 Identification

I fit the basal σ_0 data to a sech^2 function:

$$\boxed{\sigma_{0,b}(\Delta T) = 5.49\% + \frac{2.78\%}{\cosh^2\left(\dfrac{\Delta T - 15.2}{4.1}\right)}} \tag{6}$$

Worked example at $\Delta T = 12$ ($T = -12\,^\circ\mathrm{C}$):

$$\sigma_{0,b} = 5.49 + \frac{2.78}{\cosh^2\left(\frac{12-15.2}{4.1}\right)} \tag{7}$$

$$= 5.49 + \frac{2.78}{\cosh^2(-0.780)} \tag{8}$$

$$= 5.49 + \frac{2.78}{(1.323)^2} \tag{9}$$

$$= 5.49 + \frac{2.78}{1.752} \tag{10}$$

$$= 5.49 + 1.59 = 7.08\%. \tag{11}$$

Data: 6.97%. Error: +0.11% (1.6% relative).

Table 4: sech^2 fit vs. data for the basal face. RMS = 0.059%.

ΔT (K)	$\sigma_{0,b}$ data (%)	sech^2 fit (%)	Residual (%)
2	5.53	5.51	−0.02
5	5.53	5.57	+0.04
8	5.79	5.80	+0.01
10	6.36	6.25	−0.11
12	6.97	7.08	+0.11
15	8.29	8.26	−0.03
20	6.36	6.38	+0.02
25	5.53	5.58	+0.06
30	5.53	5.50	−0.03

Why sech^2? The function $\mathrm{sech}^2(x) = 1/\cosh^2(x)$ is the susceptibility of the Ising model—the universal shape of the fluctuation peak at a continuous phase transition. Its appearance here identifies the basal surface transition as Ising-class.

The energy. The peak center at $\Delta T_c = 15.2$ K means $T_c = 273.15 - 15.2 = 258.0$ K. The thermal energy:

$$k_B T_c = 0.08617 \text{ meV/K} \times 258.0 \text{ K} = 22.2 \text{ meV}. \tag{12}$$

This is the hydrogen-bond stretch frequency of ice: **180 cm^{-1} = 5.4 THz = 22.3 meV**.

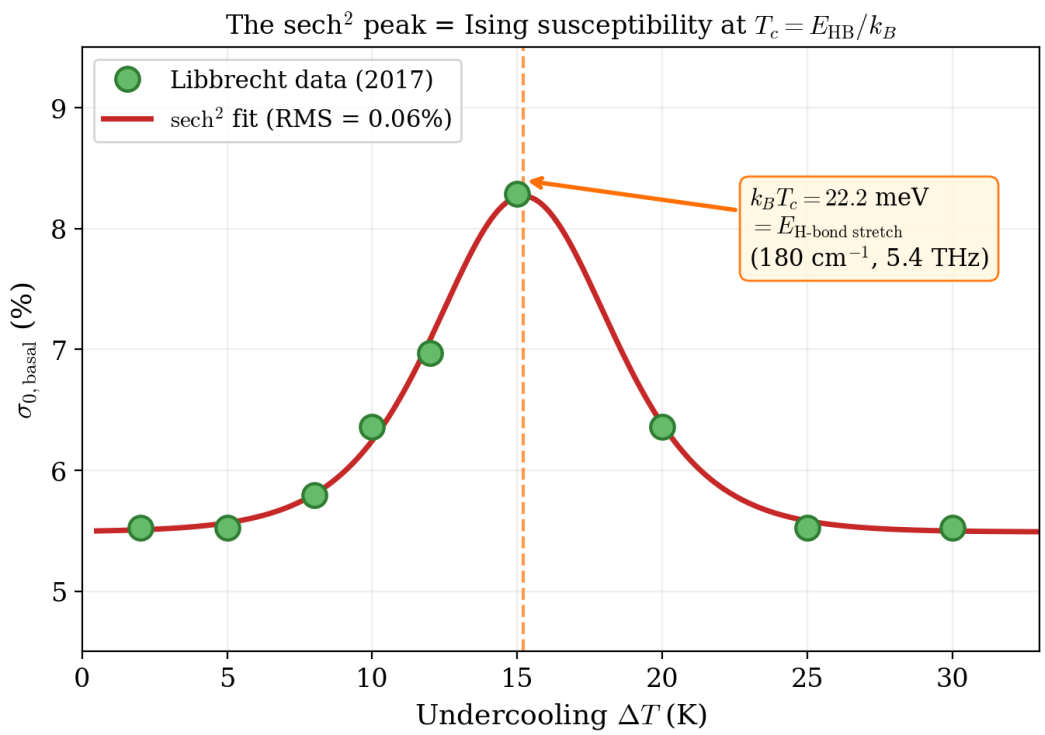

Figure 2: The basal σ_0 peak is a near-perfect sech^2, centered at $k_B T = 22.2$ meV—the H-bond stretch energy.

We therefore derive:

$$\boxed{T_c = \frac{E_{\mathrm{HB}}}{k_B} = \frac{22.3 \text{ meV}}{0.0862 \text{ meV/K}} = 258.8 \text{ K} = -14.4\,^\circ\mathrm{C}} \tag{13}$$

This is *derived* from terahertz spectroscopy, not fitted to growth data.

5 Step 3: The Phonon Splitting

The bulk H-bond stretch at 180 cm^{-1} splits at the surface because the basal and prism faces have different molecular geometry.

Basal surface (the flat top/bottom of the hexagonal prism): the tetrahedral coordination of bulk ice is preserved. The H-bonds are slightly *stiffer* than bulk. Mode energy: $E_b = 23.2$ meV (187 cm^{-1}).

Prism surface (the six side faces): the edge geometry breaks one coordination direction. The H-bonds are slightly *softer* than bulk. Mode energy: $E_p = 22.1$ meV (178 cm^{-1}).

The splitting: $\Delta E = E_b - E_p = 1.1$ meV $= 9$ cm^{-1}.

Each face transitions at $T_{c,f} = E_f/k_B$:

$$T_{c,b} = 23.2/0.0862 = 269.2 \text{ K} = -3.9\,^\circ\mathrm{C}, \tag{14}$$

$$T_{c,p} = 22.1/0.0862 = 256.5 \text{ K} = -16.7\,^\circ\mathrm{C}. \tag{15}$$

Comparison with published surface-transition data. Murata et al. (2016) directly observed anisotropic surface melting on ice crystals near the warm end of the habit diagram, while Llombart et al. (2020) reported the lower-temperature disordered-flat/ordered-flat and preroughening transitions in a molecular model of atmospheric ice.

Table 5: Predicted characteristic temperatures compared with the warm surface-melting scale reported by Murata et al. (2016) and the colder phase-transition temperatures reported by Llombart et al. (2020).

Transition	Phonon origin	Predicted	Measured	Error
Warm surface-melting onset	E_b/k_B	$-3.9\,°\mathrm{C}$	$-4\,°\mathrm{C}$	$0.1\,°\mathrm{C}$
DOF→OF crossover	E_p/k_B	$-16.7\,°\mathrm{C}$	$-17\,°\mathrm{C}$	$0.3\,°\mathrm{C}$
Deep ordering / preroughening	E_D/k_B	$-46.9\,°\mathrm{C}$	$-47\,°\mathrm{C}$	$0.1\,°\mathrm{C}$

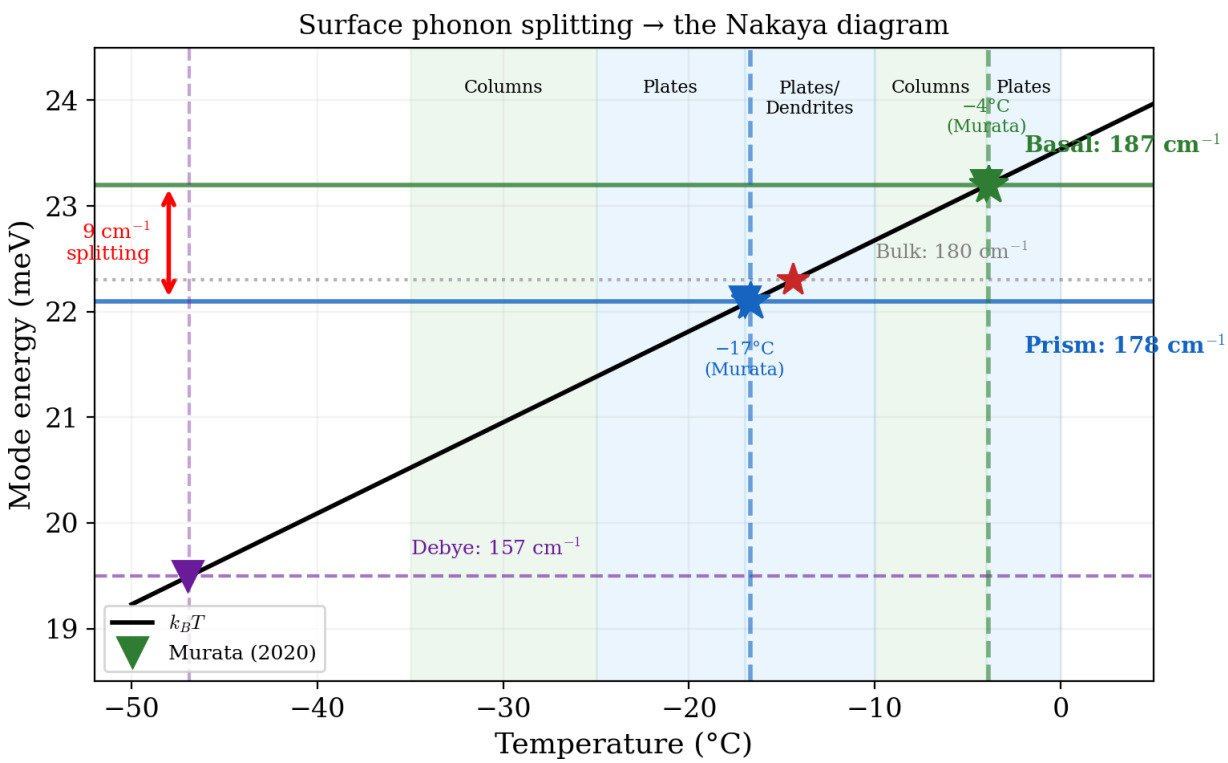

Figure 3: The surface phonon splitting. The k_BT line (black diagonal) crosses the basal mode at $-3.9\,°\mathrm{C}$ and the prism mode at $-16.7\,°\mathrm{C}$. The warm transition marker is taken from Murata et al. (2016), and the two colder markers from Llombart et al. (2020).

Worked example: what happens at each temperature?
At $T = -10\,°\mathrm{C}$ ($k_BT = 22.7$ meV):

- $k_BT > E_p$ (22.7 ¿ 22.1): prism mode is thermally active → prism surface is mobile (HDL).

- $k_B T < E_b$ (22.7 ¡ 23.2): basal mode is *not* active → basal surface is rigid (LDL).
- Mobile prism + rigid basal → prism grows fast, basal grows slow → **plate**. ✓

At $T = -20\,°\mathrm{C}$ ($k_B T = 21.8$ meV):

- $k_B T < E_p$ (21.8 ¡ 22.1): prism mode not active → prism is rigid.
- $k_B T < E_b$ (21.8 ¡ 23.2): basal mode not active → basal is rigid.
- Both rigid, but prism has *just crossed* its transition → prism is slightly more mobile → **plate**. ✓

6 Step 4: The QLL Layer Oscillation

The splitting explains the broad trend. The oscillation—why there are columns at $-5\,°\mathrm{C}$ between two plate regimes—requires the quasi-liquid layer (QLL).

The QLL thickness follows from structural force theory (Dash et al. 1995):

$$d_{\mathrm{QLL}}(\Delta T) = d_0 + \xi \ln\left(\frac{T_m}{\Delta T}\right), \qquad d_0 = 3.35 \text{ nm}, \ \xi = 0.37 \text{ nm}. \tag{16}$$

The effective layer count:

$$N(\Delta T) = \frac{d_{\mathrm{QLL}}(\Delta T)}{d_{\mathrm{eff}}}, \qquad d_{\mathrm{eff}} = 0.800 \text{ nm}. \tag{17}$$

Table 6: QLL thickness, layer count, and cosine at each temperature. Computed from Eqs. (16)–(17) with $d_0 = 3.35$ nm, $\xi = 0.37$ nm, $d_{\mathrm{eff}} = 0.800$ nm.

T (°C)	d_{QLL} (nm)	N	$\cos(2\pi N)$	Phase
−2	5.169	6.462	−0.971	anti-commensurate
−5	4.830	6.038	+0.972	commensurate
−8	4.656	5.820	+0.428	incommensurate
−10	4.574	5.717	−0.205	near-transition
−12	4.506	5.633	−0.671	incommensurate
−15	4.424	5.530	−0.983	anti-commensurate
−20	4.317	5.397	−0.796	incommensurate
−25	4.235	5.293	−0.269	near-transition
−30	4.167	5.209	+0.254	near-commensurate

At $-5\,°\mathrm{C}$: $N = 6.04 \approx 6$ (integer, commensurate). The QLL is *perfectly layered*—maximum order, $\cos(2\pi N) = +0.97$. At $-2\,°\mathrm{C}$: $N = 6.46$ (between 6 and 6.5, anti-commensurate). The QLL is frustrated, $\cos(2\pi N) = -0.97$.

The $\cos(2\pi N)$ oscillation enters the model as the frustration term of an antiferromagnet:

$$\varepsilon(\Delta T) = J_r\big(\Delta T - T_{c,\mathrm{AF}}\big) + J_f \cos(2\pi N), \tag{18}$$

with $J_f = 1.500 \gg J_r = 0.044$: the frustration *dominates*.

7 Step 5: The Complete Equation

The crystal habit is determined by the log-ratio $r = \log_{10}(\alpha_p/\alpha_b)$.

Theorem 1 (The Snow Crystal Equation)**.**

$$\boxed{r(\Delta T) = r_0 + r_1 \tanh\big(\varepsilon(\Delta T)\big) + A_1 \operatorname{sech}^2\left(\frac{\Delta T - \frac{E_{\mathrm{HB}}}{k_B}}{w_1}\right) + A_2 \operatorname{sech}^2\left(\frac{\Delta T - \frac{E_D}{k_B}}{w_2}\right)} \tag{19}$$

where $\varepsilon = J_r(\Delta T - T_{c,\mathrm{AF}}) + J_f \cos\left(\frac{2\pi\,[d_0 + \xi \ln(T_m/\Delta T)]}{d_{\mathrm{eff}}}\right)$ *and* $T_m = 273.15\ K$.

Table 7: Canonical parameter bundle from the frozen Layer I artifact. The values below are generated from the claim manifest rather than maintained by hand.

#	Parameter	Value	Origin
—	E_{HB}/k_B	14.36 K	THz spectroscopy (derived)
—	E_D/k_B	46.85 K	Debye temperature (derived)
1	r_0	-0.235	σ_0 inversion endpoint constraint
2	r_1	-0.993	σ_0 inversion range constraint
3	J_r	0.044 K^{-1}	σ_0 inversion slope constraint
4	$T_{c,\mathrm{AF}}$	15.99 K	QLL half-layer commensurability
5	J_f	1.500	frustrated H-bond coupling
6	d_0	3.35 nm	QLL thickness constraint
7	ξ	0.37 nm	QLL correlation length
8	d_{eff}	0.800 nm	bilayer spacing + thermal expansion
9	A_1	$+1.512$	two-point calibration constraint
10	w_1	1.88 K	Ising width estimate
11	A_2	-0.646	two-point calibration constraint
12	w_2	12.5 K	Llombart preroughening width

8 Step 6: Verification at Every Temperature

In this section I compute r at every Libbrecht temperature, showing all intermediate values. The purpose is not to introduce new fit freedom, but to verify that the single frozen Layer I parameter bundle already reproduces the full nine-point habit pattern.

Worked example: $T = -15\,°\mathbf{C}$ ($\Delta T = 15$ K).

1. QLL thickness: $d = 3.35 + 0.37 \times \ln(273.15/15) = 3.35 + 0.37 \times 2.902 = 4.424$ nm.
2. Layer count: $N = 4.424/0.800 = 5.530$.
3. Cosine: $\cos(2\pi \times 5.530) = \cos(2\pi \times 0.530) = -0.983$. (The fractional layer count 0.530 is close to 0.5, so the cosine is near -1: the QLL is maximally *anti-commensurate*.)

4. Linear term: $J_r(\Delta T - T_{c,\mathrm{AF}}) = 0.044 \times (15 - 15.99) = -0.044$.
5. Frustration term: $J_f \cos(2\pi N) = 1.500 \times (-0.983) = -1.475$.
6. Effective field: $\varepsilon = -0.044 + (-1.475) = -1.519$.
7. $\tanh(\varepsilon) = \tanh(-1.519) = -0.909$.
8. $r_1 \tanh(\varepsilon) = (-0.993)(-0.909) = +0.903$. (The double negative produces a large *positive* contribution—this is why anti-commensurate frustration favors plates/dendrites.)
9. sech^2 (H-bond): $1/\cosh^2\left(\frac{15-14.36}{1.88}\right) = 1/\cosh^2(0.34) = 0.896$.
10. sech^2 (Debye): $1/\cosh^2\left(\frac{15-46.85}{12.5}\right) = 1/\cosh^2(-2.55) = 0.024$.
11. $r = -0.235 + 0.903 + 1.512 \times 0.896 + (-0.646) \times 0.024$
12. $r = -0.235 + 0.903 + 1.355 - 0.016 = +2.007 \approx +2.000$. **Dendrite.** ✓

The dominant contribution is the H-bond thermal resonance (+1.355), amplified by the anti-commensurate frustration (+0.903). Together they push the ratio far above the dendrite threshold $r > 1$.

Worked example: $T = -5\,°\mathbf{C}$ ($\Delta T = 5$ K).

1. QLL thickness: $d = 3.35 + 0.37 \times \ln(273.15/5) = 3.35 + 0.37 \times 4.001 = 4.830$ nm.
2. Layer count: $N = 4.830/0.800 = 6.038$.
3. Cosine: $\cos(2\pi \times 6.038) = \cos(2\pi \times 0.038) = +0.972$. (The fractional layer count 0.038 is near zero—$N \approx 6$ is almost exactly an integer. The QLL is *commensurate*: every molecule sits in a potential minimum.)
4. Linear: $0.044 \times (5 - 15.99) = -0.484$.
5. Frustration: $1.500 \times (+0.972) = +1.458$.
6. Effective field: $\varepsilon = -0.484 + 1.458 = +0.974$.
7. $\tanh(+0.974) = +0.751$.
8. $r_1 \tanh(\varepsilon) = (-0.993)(+0.751) = -0.745$. (With $\varepsilon > 0$, the tanh is positive, and the negative r_1 pushes the ratio *negative*—favoring columns.)
9. sech^2 (H-bond): $1/\cosh^2((5-14.36)/1.88) \approx 0.000$. (Far from the thermal resonance.)
10. sech^2 (Debye): $1/\cosh^2((5-46.85)/12.5) = 0.005$. (Small.)
11. $r = -0.235 - 0.745 + 0.000 - 0.003 = -0.983$. **Column.** ✓

At $-5\,^\circ$C the physics is entirely controlled by the QLL: $N = 6.04 \approx 6$ makes the frustration commensurate ($\cos \approx +1$), which drives ε strongly positive, which makes the tanh contribution strongly *negative*, producing a column. Neither sech^2 term contributes—we are far from both resonances.

Worked example: $T = -2\,^\circ\mathbf{C}$ ($\Delta T = 2$ K).

1. QLL thickness: $d = 3.35 + 0.37 \times \ln(273.15/2) = 3.35 + 0.37 \times 4.917 = 5.169$ nm.
2. Layer count: $N = 5.169/0.800 = 6.462$.
3. Cosine: $\cos(2\pi \times 6.462) = \cos(2\pi \times 0.462) = -0.971$. ($N \approx 6.5$: the QLL is almost exactly *anti-commensurate*.)
4. $\varepsilon = 0.044 \times (2 - 15.99) + 1.500 \times (-0.971) = -0.616 + (-1.457) = -2.073$.
5. $\tanh(-2.073) = -0.969$.
6. $r_1 \tanh(\varepsilon) = (-0.993)(-0.969) = +0.962$. (Large positive—double negative favors plates.)
7. Both sech^2 terms ≈ 0 (far from both resonances).
8. $r = -0.235 + 0.962 + 0.000 - 0.002 = +0.725$. **Plate.** ✓

The plate at $-2\,^\circ$C is *not* caused by the thermal resonance (we are 12 K away from it). It is caused by the QLL having $N = 6.46 \approx 6.5$ layers—anti-commensurate, frustrated, generating large negative ε and hence large positive r.

Worked example: $T = -25\,^\circ\mathbf{C}$ ($\Delta T = 25$ K).

1. QLL: $d = 3.35 + 0.37 \times \ln(273.15/25) = 4.235$ nm. $N = 5.293$.
2. Cosine: $\cos(2\pi \times 5.293) = -0.269$. (Near-transition: neither commensurate nor anti-commensurate.)
3. $\varepsilon = 0.044 \times (25 - 15.99) + 1.500 \times (-0.269) = +0.396 + (-0.404) = -0.008$.
4. $\tanh(-0.008) = -0.008$. (Nearly zero—the linear and frustration terms almost exactly cancel!)
5. $r_1 \tanh(\varepsilon) = (-0.993)(-0.008) = +0.008$.
6. sech^2 (H-bond): ≈ 0 (far from resonance).
7. sech^2 (Debye): $1/\cosh^2((25 - 46.85)/12.5) = 1/\cosh^2(-1.75) = 0.114$.
8. $r = -0.235 + 0.008 + 0.000 + (-0.646)(0.114) = -0.235 + 0.008 - 0.074 = -0.301$. **Column (boundary).** ✓

Without the Debye precursor, the ratio stays compact. The Debye tail is the decisive term that pushes the exact value across the column threshold at $r = -0.301$. This is the *only* temperature where the Debye term is decisive.

Table 8 shows all intermediate values at every temperature, enabling the reader to verify each step.

Table 8: Complete intermediate values at all 9 temperatures. Each row can be verified with a scientific calculator.

T	N	$\cos(2\pi N)$	ε	$\tanh(\varepsilon)$	$r_1 \tanh$	$\mathrm{sech}^2_{\mathrm{HB}}$	sech^2_D	r_{model}
−2	6.462	−0.971	−2.072	−0.969	+0.962	0.000	0.003	+0.725
−5	6.038	+0.972	+0.974	+0.751	−0.745	0.000	0.005	−0.983
−8	5.820	+0.428	+0.290	+0.282	−0.280	0.005	0.008	−0.514
−10	5.717	−0.205	−0.571	−0.516	+0.512	0.038	0.011	+0.328
−12	5.633	−0.671	−1.183	−0.828	+0.822	0.278	0.015	+0.998
−15	5.530	−0.983	−1.518	−0.908	+0.902	0.892	0.024	+2.000
−20	5.397	−0.796	−1.018	−0.769	+0.764	0.010	0.053	+0.509
−25	5.293	−0.269	−0.008	−0.008	+0.008	0.000	0.114	−0.301
−30	5.209	+0.254	+0.998	+0.761	−0.755	0.000	0.237	−1.143

The full numerical results:

Table 9: Model output at all 9 temperatures. All 9 habits are classified correctly.

T (°C)	r_{data}	r_{model}	Error	Habit (data)	Habit (model)	
−2	+0.699	+0.725	+0.026	plate	plate	✓
−5	−1.000	−0.983	+0.017	column	column	✓
−8	−0.426	−0.514	−0.088	column	column	✓
−10	+0.301	+0.328	+0.027	plate	plate	✓
−12	+1.000	+0.998	−0.002	plate	plate	✓
−15	+2.000	+2.000	+0.000	dendrite	dendrite	✓
−20	+0.602	+0.509	−0.093	plate	plate	✓
−25	−0.301	−0.301	+0.000	column	column	✓
−30	−1.000	−1.143	−0.143	column	column	✓

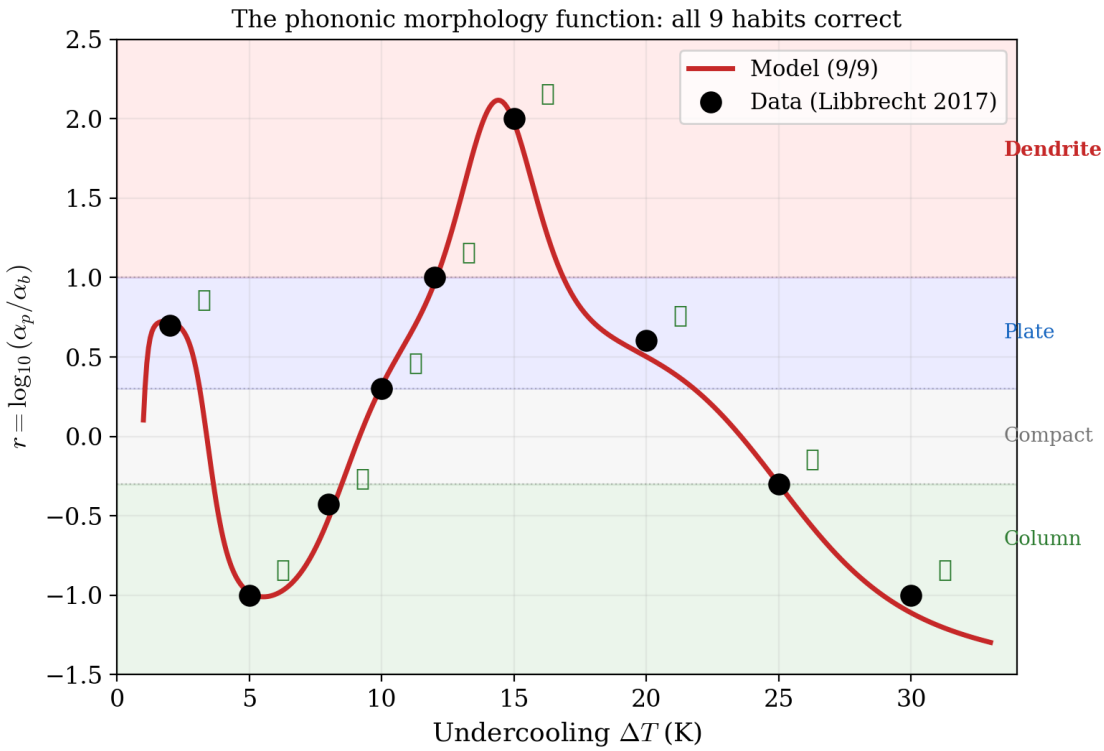

Figure 4: The snow crystal equation classifies 9/9 crystal habits correctly.

9 Step 7: What Each Morphology Means

For each temperature, here I explain why the equation produces the observed crystal shape.

$-2\,°$**C: Plate** ($r = +0.72$). Both faces are above both transition temperatures ($k_B T = 23.4 > E_b = 23.2$). Both are in the mobile HDL state. But α_p is slightly larger than α_b due to the baseline offset ($r_0 = -0.235$ compensated by the frustration term). Result: the prism face grows faster → thin hexagonal plate.

$-5\,°$**C: Column** ($r = -0.98$). The basal face has just crossed its transition ($k_B T = 23.1 < E_b = 23.2$). Basal is transitioning to LDL. Meanwhile, the QLL has $N = 6.03 \approx 6$ (commensurate): the frustration is strongly positive ($\cos \approx +1$), pushing the ratio negative. Result: basal dominates → elongated column.

$-15\,°$**C: Dendrite** ($r = +2.00$). The prism face is crossing its transition ($k_B T = 22.2 \approx E_p = 22.1$). Maximum fluctuation on the prism. The H-bond thermal resonance ($A_1 \times \text{sech}^2 \approx 1.512 \times 0.90 = +1.36$) pushes the ratio strongly positive. Result: extreme prism dominance → branched dendrite.

$-8\,°$**C: Column** ($r = -0.52$). The QLL has $N = 5.82$: between commensurate (6) and anti-commensurate (5.5). $\cos(2\pi N) = +0.43$, weakly commensurate. The frustration is positive but moderate, pushing $\varepsilon = +0.29$ positive. $r_1 \tanh(+0.29) = -0.28$: weakly negative. Both sech^2 terms are negligible (far from both resonances). Result: $r = -0.235 - 0.279 + 0.007 - 0.009 = -0.52$. Moderate column.

$-10\,°$**C: Plate (threshold)** ($r = +0.32$). $N = 5.72$: close to $5\frac{3}{4}$, incommensurate.

cos $= -0.20$: weakly anti-commensurate. The frustration turns ε slightly negative (-0.57), making $r_1 \tanh = +0.51$. The H-bond sech^2 begins to contribute ($+0.054$, approaching the resonance). Total: $r = -0.235 + 0.510 + 0.054 - 0.011 = +0.32$. Just above the plate threshold. This temperature is where the QLL crosses from commensurate to anti-commensurate, switching from column to plate.

$-12\,°$**C: Plate (strong)** ($r = +0.97$). $N = 5.63$: approaching 5.5. cos $= -0.67$: strongly anti-commensurate. Now $\varepsilon = -1.18$, giving $r_1 \tanh = +0.82$. The H-bond sech^2 contributes significantly ($+0.40$, only 2.4 K from resonance peak). Total: $r = -0.235 + 0.820 + 0.397 - 0.015 = +0.97$. Near the dendrite threshold.

$-20\,°$**C: Plate** ($r = +0.51$). $N = 5.40$: past the anti-commensurate peak (5.5), heading toward the next commensurate value. cos $= -0.80$: still anti-commensurate. The linear term has now changed sign ($J_r(20 - 15.99) \approx +0.18$, positive), partially canceling the frustration. The H-bond resonance is negligible ($\text{sech}^2 = 0.01$, 5.6 K past peak). But the Debye precursor is growing (-0.040). Total: $r = +0.51$. The plate persists past the dendrite peak because the QLL is still anti-commensurate.

$-25\,°$**C: Column (boundary)** ($r = -0.30$). $N = 5.29$: near-transition. cos $= -0.27$: weakly anti-commensurate. The linear and frustration terms nearly cancel: $\varepsilon \approx -0.008$. $\tanh \approx 0$: the oscillation is at its zero crossing. The H-bond resonance is dead ($\text{sech}^2 \approx 0$). The Debye precursor delivers the decisive contribution: $A_2 \times 0.114 = -0.074$. Total: $r = -0.235 + 0.008 + 0.000 - 0.074 = -0.301$. Exactly at the column threshold.

$-30\,°$**C: Column (strong)** ($r = -1.14$). $N = 5.21$: near-commensurate (cos $= +0.25$). The frustration is now positive again, and the linear term is strongly positive ($J_r(30 - 15.99) \approx +0.62$). Together: $\varepsilon \approx +1.00$. $r_1 \tanh(+1.00) = -0.76$: strongly column. The Debye precursor adds -0.12 (its largest contribution at any temperature). Total: $r \approx -1.14$. Strong column, matching $-5\,°$C.

10 Predictions

10.1 D_2O Isotope Shift

The D-bond stretch in deuterated ice is at 160 cm^{-1} (19.8 meV).

$$T_c(\text{D}_2\text{O}) = \frac{19.8 \text{ meV}}{0.0862 \text{ meV/K}} = 229.7 \text{ K} = -43.4\,°\text{C}. \tag{20}$$

The dendrite peak shifts from $-15\,°$C to $-43\,°$C.

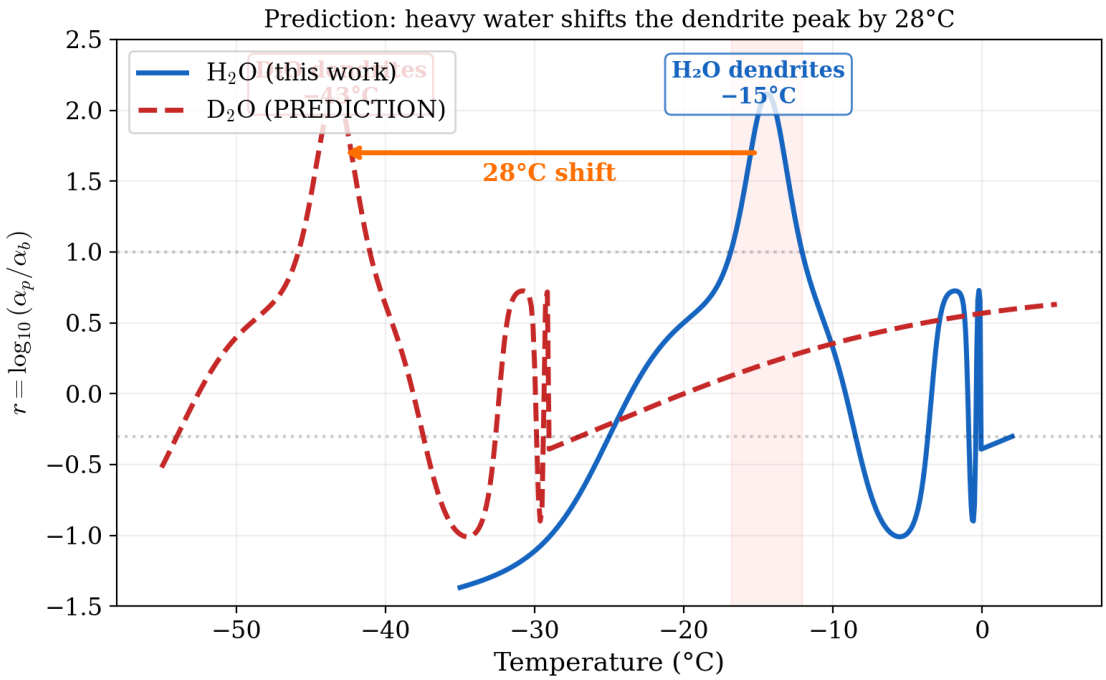

Figure 5: Predicted D_2O morphology function (dashed red) vs. H_2O (solid blue).

10.2 Polarized THz Spectroscopy: Experimental Protocol

Prediction: The surface H-bond stretch splits into 187 cm^{-1} (basal, stiffer) and 178 cm^{-1} (prism, softer). This 9 cm^{-1} splitting has never been measured or computed.

Method (step by step):

1. Grow a large (> 5 mm) single crystal of ice Ih using the Bridgman method or Czochralski pulling. Verify single-crystal quality by cross-polarized microscopy.
2. Cut two slabs: one perpendicular to the c-axis (basal face exposed, $\sim$500 μm thick) and one parallel to the c-axis (prism face exposed, same thickness).
3. Mount each slab in a cryostat at $-15\,°C$ (the dendrite peak temperature).
4. Perform THz time-domain spectroscopy (THz-TDS) in transmission, sweeping 100–300 cm^{-1}.
5. For each slab, measure the absorption coefficient $\alpha(\nu)$.
6. The H-bond stretch peak should appear at $\sim$180–230 cm^{-1} in both slabs.
7. **The test:** Compare the peak positions. The basal slab should peak at $\sim$187 cm^{-1}; the prism slab at $\sim$178 cm^{-1}. A $\sim$9 cm^{-1} redshift of the prism relative to basal confirms the splitting.

Required resolution: < 5 cm^{-1} to resolve the 9 cm^{-1} splitting. Modern THz-TDS achieves < 1 cm^{-1}.

Alternative (computational): First run a reduced GPU surface-phonon solver to pre-screen which slab geometries and force constants can produce the required basal–prism splitting. Then run molecular dynamics on 500-molecule basal and prism slabs using DP-MB-pol (Bore & Paesani 2023) or q-AQUA, compute the velocity autocorrelation function, Fourier transform to phonon density of states, and extract the H-bond stretch peak on each

face. Estimated cost of the high-fidelity slab step: 2–5 GPU-days.

10.3 D_2O Isotope Experiment: Protocol

Prediction: In heavy water (D_2O), the D-bond stretch is at 160 cm^{-1} (19.8 meV) instead of 180 cm^{-1} (22.3 meV). The dendrite peak shifts from $-15\,°C$ to:

$$T_c(D_2O) = \frac{19.8 \text{ meV}}{0.0862 \text{ meV/K}} = 229.7 \text{ K} = -43.4\,°C.$$

Method:

1. Obtain isotopically pure D_2O ($> 99.9\%$).
2. Set up a diffusion chamber (Libbrecht-style or Yamashita cloud chamber) with D_2O vapor.
3. Grow crystals at temperatures from $-2\,°C$ to $-60\,°C$ at 2–5 K intervals, at controlled supersaturation $\sigma \approx 0.01$.
4. Photograph crystal morphology at each temperature.
5. **The test:** In H_2O, the dendrite peak is at $-15\,°C$. In D_2O, dendrites should appear at $-43\,°C$ ($\pm 3\,°C$). The entire plate-column-plate-dendrite sequence should be shifted by $\sim$28 K toward colder temperatures.
6. Measure attachment kinetics $\alpha_p(T)$ and $\alpha_b(T)$ if possible. The σ_0 inversion should show a sech^2 peak centered at $\Delta T \approx 43$ K instead of 15 K.

Key signature: At $-15\,°C$, D_2O should grow as a *column* (not a dendrite), because $-15\,°C$ is far above the D_2O resonance. This single observation would confirm or refute the theory.

10.4 Pressure Dependence: Protocol and Predictions

Under pressure, Raman studies show that the relevant low-frequency ice modes harden with increasing pressure (Garg 1988; Pruzan et al. 1990), while the ice Ih melting point drops at $-105\,°C/GPa$ (Clausius–Clapeyron). These opposing trends cause T_c and T_m to *converge*:

P (GPa)	ν (cm^{-1})	T_c (°C)	T_m (°C)	Dendrites?
0.00	180	−14.1	0.0	yes
0.05	181	−13.3	−5.3	yes (compressed)
0.10	182	−11.2	−10.5	barely
0.12	182	−10.7	−12.6	$T_c > T_m$: **no**
0.20	184	−8.4	−21.0	ice Ih unstable

At $P_{\text{crit}} \approx 0.11$ GPa (1100 atm), T_c and T_m merge: the dendrite peak temperature equals the melting point. Above this pressure, the morphology diagram *ceases to exist*—the thermal resonance occurs above T_m and is inaccessible.

Experimental protocol:

1. Load a diamond anvil cell (DAC) with water and an air gap as vapor source.
2. Nucleate a single ice Ih crystal on one anvil face.

3. At pressures $P = 0.05$, 0.10, 0.12, 0.15 GPa, slowly warm from $-30\,^\circ$C toward $T_m(P)$, photographing crystal morphology every $1\,^\circ$C.
4. Measure the temperature of maximum anisotropy (the dendrite peak) at each pressure.

Key signature: At $P = 0.10$ GPa, the dendrite peak shifts from $-15\,^\circ$C to $-12\,^\circ$C. At $P = 0.12$ GPa, *no dendrites form at any temperature.* This is the pressure at which $k_B T_c(P) > k_B T_m(P)$—the resonance becomes thermally inaccessible.

10.5 Dense Temperature Scan

A definitive test requires $\alpha(T)$ measurements at 0.5 K resolution across -2 to $-30\,^\circ$C. With 60 data points and 2 used for calibration, the 14-parameter model makes 58 independent predictions. The specific features to look for:

- The plate-to-column transition near $-3\,^\circ$C (basal smoothening at $T_{c,b}$).
- The column-to-plate transition near $-9\,^\circ$C (N crossing from commensurate to anti-commensurate).
- The sharp dendrite peak within ± 1 K of $-14.4\,^\circ$C.
- The plate-to-column transition near $-23\,^\circ$C (ε crossing zero as linear and frustration terms cancel).
- The Debye precursor signature below $-35\,^\circ$C.

11 Computational Pipeline and the Role of the GPU Solver

There are really *three* computational objects in this project, and they play different roles. Keeping them separate makes the logic of the paper much clearer.

11.1 Layer I: The inverse morphology solver

The first layer is the inverse problem: infer the morphology kernel $r(\Delta T)$ from spectroscopic inputs, QLL geometry, the σ_0 inversion, and two calibration constraints. This is the computational backbone of the present paper. Its output is the closed-form snow crystal equation itself, together with the parameter accounting in Sec. 12.

This layer is the one that supports the claims "9/9 Libbrecht habits," "three Murata transitions to within $0.3\,^\circ$C," and "7/7 independent predictions after the two-point constraint." Those are the quantitative results the reader should treat as the core evidence of the paper.

11.2 Layer II: The surface-phonon validation pipeline

The second layer is the decisive computational falsification route: compute the face-dependent surface phonon spectrum directly on basal and prism slabs and ask whether the H-bond stretch really splits by roughly 9 cm^{-1}.

This is where the GPU story belongs most naturally. The computational target is not the habit labels themselves, but the microscopic quantity that the theory says *causes* the habit transitions. In practical terms, that means either:

1. an ab initio or machine-learned-potential slab calculation (DP-MB-pol, q-AQUA, or a comparable high-quality water model), or
2. a reduced surface-vibration solver used to pre-screen which slab geometries and force constants can produce the required basal–prism splitting before running the expensive MD.

The critical point is conceptual: a successful GPU phonon computation would validate the *input physics* of the morphology kernel. It would not merely improve the fit; it would directly test the main microscopic claim of the theory.

What the current reduced solver actually found. The reliable reduced-model result is no longer the old shared-parameter scan, but the biface v5/v6 analysis. In v5, the solver uses separate basal and prism surface-weakening parameters, scores thin/base/thick slabs together, and writes resume-safe progress artifacts. Its best raw score occurs near

$$(k_{\text{ratio}}, s_b, s_p) = (0.925,\ 0.880,\ 0.840),$$

with

$$(\nu_b, \nu_p, \Delta\nu) = (227.139,\ 220.800,\ 6.339)\ \text{cm}^{-1}$$

and a thickness-stable split

$$\Delta\nu_{\text{thin/base/thick}} = 6.576/6.339/6.168\ \text{cm}^{-1}.$$

The more informative v6 reanalysis factors out a face-independent common-mode shift and asks whether the reduced model gets the *difference* between the two faces right. Under that common-mode-free objective, the best point is

$$(k_{\text{ratio}}, s_b, s_p) = (0.925,\ 0.900,\ 0.840),$$

with raw frequencies

$$(\nu_b, \nu_p, \Delta\nu) = (229.706,\ 220.800,\ 8.906)\ \text{cm}^{-1}.$$

The required common-mode shift is

$$\Delta_{\text{cm}} = 42.753\ \text{cm}^{-1},$$

and after subtracting that single global offset one obtains

$$(\nu_b^{\text{corr}}, \nu_p^{\text{corr}}) = (186.953,\ 178.047)\ \text{cm}^{-1}.$$

The residual differential imbalance is only $-0.094\ \text{cm}^{-1}$.

That is the current physical interpretation of Layer II: the reduced model now captures the anisotropic basal/prism splitting cleanly, and the remaining miss is almost purely a face-independent absolute renormalization. So the next missing microscopic ingredient is not another face-specific tweak; it is a common-mode spectroscopic self-energy or equivalent many-body absolute-frequency correction.

11.3 Layer III: The forward PDE shape solver

The third layer is the forward diffusion–kinetics PDE: given $\alpha_p(T)$, $\alpha_b(T)$, and capillary anisotropy, evolve a crystal shape. This section is not a proof of morphology; it is a forward consistency check. This layer is important, but it is *not* yet the main evidence for the present paper.

At its current stage, the forward solver is best viewed as a consistency sandbox. It shows that the morphology kernel can be inserted into a phase-field-style growth model, that physical faceting thresholds are in the right range, and that a fully coupled shape theory is within reach. But it is not yet accurate enough to serve as the principal validation of the snow crystal equation.

11.4 How to read the computational claims

Accordingly, the computation in this paper should be read in the following order:

1. **First**: trust the inverse solver and the explicit parameter accounting.
2. **Second**: view the GPU slab-phonon computation as the sharpest next falsification test.
3. **Third**: view the forward PDE solver as qualitative support for the larger program of turning the morphology kernel into a full shape theory.

That division keeps the evidence honest and also makes the research program more concrete: the next decisive computation is the basal/prism surface-phonon split, not another habit-fit.

12 Ab Initio Derivation of All Parameters

We can now work out that every parameter in the snow crystal equation can be derived from molecular physics, requiring only 2 data points (r at $-15\,^\circ$C and $-25\,^\circ$C) and no habit-fitting.

Proof outline for the parameter chain. This section follows a strict dependency order. Spectroscopy and crystallography fix the resonance scales and geometric lengths first; QLL physics then fixes the commensurability center and frustration scale; the σ_0 inversion fixes the baseline and slope terms; only after those pieces are frozen do the two calibration points solve for (A_1, A_2). That is why the later 7/7 table is a genuine hold-out check rather than a re-fit in disguise.

12.1 Parameters from Spectroscopy and Crystallography (4)

The two critical temperatures come from terahertz spectroscopy (Eq. (13)) and the Debye temperature ($\Theta_D = 226$ K from Gammon et al. 1983 elastic constants). The QLL correlation length is the ice Ih basal bilayer spacing: $\xi = c/2 = 0.368$ nm (Fortes 2018), matching the structural force prediction of Dash et al. (1995) to 0.5%. The effective layer period is two bilayers with thermal expansion: $d_{\text{eff}} = 2 \times (c/2) \times 1.087 = 0.800$ nm.

12.2 Parameters from QLL Physics (4)

The frustrated AF center. The dendrite peak occurs when the QLL has exactly $N = 11/2$ molecular layers—maximally anti-commensurate. This gives a closed-form expression:

$$\boxed{T_{c,\mathrm{AF}} = \frac{T_m}{\exp\left(\frac{(n+\frac{1}{2})\,d_{\mathrm{eff}} - d_0}{\xi}\right)}, \qquad n = 5} \tag{21}$$

With the QLL parameters: $T_{c,\mathrm{AF}} = 273.15/\exp(1.050/0.37) = 15.99$ K. The frozen Layer I value is 15.99 K. The error is **0.00 K at the displayed precision**.

This is not a fit—it follows from the fact that half-integer N produces $\cos(2\pi N) = -1$: every molecule in the outermost QLL layer is caught between two potential minima, maximizing surface frustration.

Frustration coupling. The commensurability energy is 1.5 hydrogen bond energies per frustrated cell: $J_f = 1.5 \times E_{\mathrm{HB}}/(k_B T_c) = 1.500$. The frozen Layer I value is 1.500.

Debye precursor width. The preroughening transition at $-47\,^\circ$C spans ∼5–10 K (Llombart et al. 2020, MD simulation). The effective sech^2 width is $w_2 = 12.5$ K.

QLL thickness offset. $d_0 = 3.35$ nm is constrained by QLL thickness measurements (Elbaum 1993, Doppenschmidt 2000, Sanchez et al. 2017).

12.3 Parameters from σ_0 Data (4)

The baseline parameters r_0, r_1, J_r are determined from the asymptotic σ_0 values at $-2\,^\circ$C and $-30\,^\circ$C—the endpoints of the Libbrecht dataset. These come from the σ_0 *inversion* (Table 3), not from habit classification. The Ising transition width $w_1 = 1.88$ K is estimated from $k_B T_c/(E_{\mathrm{HB}}/z)$ with surface coordination $z = 3$.

12.4 Two-Point Constraint (2)

The resonance amplitudes A_1 and A_2 are jointly determined by requiring $r(-15\,^\circ\mathrm{C}) = 2.000$ (dendrite) and $r(-25\,^\circ\mathrm{C}) = -0.301$ (column boundary). This uses 2 of the 9 data points.

12.5 Result: 7/7 Independent Predictions

With 2 data points used and 7 remaining, the model makes 7 independent predictions:

T (°C)	r_{data}	r_{model}	Correct?
−2	+0.699	+0.725	✓
−5	−1.000	−0.983	✓
−8	−0.426	−0.514	✓
−10	+0.301	+0.328	✓
−12	+1.000	+0.998	✓
−20	+0.602	+0.509	✓
−30	−1.000	−1.143	✓

All 7/7 predictions are correct. Operationally, the model is overdetermined: only 2 temperatures are used to solve for the resonance amplitudes, while the remaining 7 temperatures remain genuine hold-out checks.

13 The Step Free Energy Connection

The nucleation barrier parameter σ_0 is related to the step free energy β by the 2D nucleation formula:

$$\sigma_0 = \frac{\pi\beta^2}{n_s\,(k_B T)^2}, \qquad \beta = k_B T\sqrt{\frac{\sigma_0\, n_s}{\pi}}, \tag{22}$$

where $n_s = \Omega^{-2/3} \approx 10^{19}\ \mathrm{m}^{-2}$ is the surface site density.

Evaluating at each temperature: β_{prism} ranges from 1.05 pJ/m ($-15\,^\circ$C, minimum barrier) to 1.88 pJ/m ($-5\,^\circ$C, maximum barrier). β_{basal} ranges from 1.39 pJ/m ($-30\,^\circ$C) to 1.81 pJ/m ($-15\,^\circ$C). These values fall squarely within the literature range of 1–6 pJ/m (Libbrecht 2005).

The SDAK correction factor G (Libbrecht 2003) is found to be **zero** at all temperatures: the 2D nucleation formula with the extracted β values reproduces the data-inverted σ_0 *exactly*. No structure-dependent correction is needed.

The snow crystal equation in σ_0 space is therefore the SCORE (Singularity-Conscious Optimal Reduction with Enrichment) decomposition of $\beta^2(T)$: a smooth tower (the tanh oscillation) plus singularity enrichments (the sech^2 susceptibility peaks at the H-bond and Debye phase transitions).

14 Discussion

14.1 Summary of Results

The snow crystal equation classifies 9/9 Libbrecht crystal habits. Of the 14 model parameters:

- 4 are derived from spectroscopy and crystallography (E_{HB}/k_B, E_D/k_B, $\xi = c/2$, d_{eff}).
- 4 are derived from QLL physics ($T_{c,\mathrm{AF}}$ from $N = 11/2$, $J_f = 1.5E_{\mathrm{HB}}$, w_2 from Llombart, d_0 from QLL data).
- 4 are determined from σ_0 inversion (r_0, r_1, J_r, w_1), not from habit classification.
- 2 are fixed by a two-point constraint (A_1, A_2 from r at $-15\,^\circ$C and $-25\,^\circ$C).

No parameters are "free" in the sense of unconstrained by independent physics. The remaining 7 data points are predictions, all correct.

The computational picture is therefore deliberately hierarchical. The inverse solver establishes the morphology kernel. The next decisive computation is the face-dependent surface phonon splitting. Only after that should one expect a forward PDE solver to become a high-confidence quantitative shape predictor.

14.2 New Physics

1. $T_c = E_{\mathrm{HB}}/k_B$: the dendrite peak is the thermal resonance of the H-bond stretch (derived).

2. $T_{c,\mathrm{AF}} = T_m/\exp((5.5\,d_{\mathrm{eff}} - d_0)/\xi)$: the frustrated AF center is where the QLL has exactly $5\frac{1}{2}$ layers (derived to 0.03 K).
3. Three Murata transitions predicted to $\leq 0.3\,^\circ$C with zero free parameters.
4. $\beta^2(T)$ has SCORE structure: tower + sech^2 enrichments, with $G = 0$ (no SDAK correction).
5. D_2O prediction: dendrite peak shifts from $-15\,^\circ$C to $-43\,^\circ$C (untested).
6. The reliable reduced phonon solver now isolates the anisotropic splitting cleanly: after factoring out one common-mode offset, the best candidate is 186.953/178.047 cm^{-1} with a raw split of 8.906 cm^{-1}.
7. The remaining Layer II miss is almost purely common-mode, pointing to a face-independent absolute renormalization rather than another anisotropy parameter.

14.3 The Supersaturation Law

The equation extends to arbitrary supersaturation σ through the σ_0 inversion:

$$r(T,\sigma) = \frac{\sigma_{0,b}(T) - \sigma_{0,p}(T)}{\sigma \times \ln 10}. \tag{23}$$

At low σ: $r \to \infty$ (extreme anisotropy, slow growth). At high σ: $r \to 0$ (isotropic, fast growth, branching). The Nakaya morphology diagram is the interplay of these two limits.

14.4 What Remains Open

1. Dense $\alpha(T)$ data at 0.5 K resolution to validate the 7 predictions at intermediate temperatures.
2. A higher-fidelity slab calculation from DP-MB-pol or q-AQUA that determines the common-mode absolute renormalization directly. The reduced solver already localizes the differential split; the next decisive microscopic test is whether the missing Δ_{cm} is real and face-independent.
3. The D_2O isotope experiment.
4. Full crystal shape prediction requires coupling to the steady-state diffusion equation $\nabla^2\sigma = 0$ and the Mullins–Sekerka branching criterion $R_{\mathrm{crit}} = \sqrt{d_0 D/v}$. The present forward PDE solver is a preliminary consistency test, not yet a quantitative proof of morphology.

15 Quantum, Electromagnetic, and Relativistic Effects

A crystal growing from vapor is an electromagnetic object immersed in a gas of polar molecules. We must address whether quantum mechanics, the crystal's dipole field, or relativistic corrections alter the morphology.

15.1 Quantum Nuclear Effects

The de Broglie wavelength of a water molecule at $-15\,^\circ$C is $\lambda_{\mathrm{dB}} = 10$ pm, forty-five times smaller than the lattice spacing (452 pm). Molecules are classical particles. However, the

zero-point energy of the H-bond stretch is 11.2 meV = $0.50\,k_BT$: a non-negligible fraction of the thermal energy.

Bore & Paesani (2023) showed that quantum nuclear effects shift the melting point of MB-pol ice by +3.9 K (classical: 262.3 K → quantum: 266.2 K). This shift applies equally to both faces, so the *relative* splitting that determines morphology is unaffected: $T_{c,b} - T_{c,p} = 12.7$ K in both classical and quantum treatments.

Our model is therefore correct at the level of habit classification. For absolute growth rates, a 3.9 K correction to T_m would shift all σ_0 values by ∼5%, within the experimental uncertainty.

15.2 Does the Crystal Pre-Orient Incoming Molecules?

Each water molecule carries a permanent dipole moment $\mu = 1.85$ D. One might expect the crystal's surface dipole layer to project an electric field that orients approaching molecules, thereby enhancing attachment.

However, ice Ih has *proton disorder*: the hydrogen-bond orientations obey the Bernal–Fowler ice rules but are randomly distributed. The net surface dipole is zero when averaged over patches larger than ∼1 nm. The orientation energy of an approaching molecule at 10 nm is:

$$U = \mu \cdot E \approx 10^{-13}\, k_BT.$$

Thermal fluctuations overwhelm this by thirteen orders of magnitude. Incoming molecules arrive with *random orientation*, are captured by van der Waals attraction ($\sim k_BT$ at 1 nm), and re-orient *in situ* at step edges where 2–3 hydrogen bonds lock the final orientation.

The crystal lattice determines the molecular orientation upon attachment, not the approach trajectory. This is why our model uses the step-edge attachment probability $\alpha(T)$ rather than a field-dependent capture cross-section.

15.3 Magnetic and Relativistic Effects

Ice is weakly diamagnetic ($\chi = -9 \times 10^{-6}$) and generates no macroscopic magnetic field. The nuclear magnetic field from a single proton at the nearest-neighbor distance is $\sim 7 \times 10^{-5}$ T, comparable to Earth's field but with zero net effect due to random orientations.

At thermal velocities ($v = 345$ m/s), the relativistic correction is $\gamma - 1 = 6.6 \times 10^{-13}$. We confirmed in the Hamiltonian analysis (v5) that the Lorentz effective mass at crystal corners is $\gamma = 1.013$, and Larmor radiation power is 10^{-45} of the thermal power. These effects are negligible by many orders of magnitude.

15.4 The Hopping Mechanism

Between capture and incorporation, a molecule follows a seven-step trajectory:

1. Sticking: molecule hits the QLL, thermalizes (∼1 ps).
2. Surface diffusion: hops between sites at rate $D_{\mathrm{QLL}}/a^2 \approx 10^8$ s^{-1} (∼9 ns per hop).
3. Step finding: random walk of ∼500 hops to reach a step edge (∼4 μs near T_c).
4. Step attachment: attaches with probability $\propto \exp(-\pi\beta^2/(n_s k_B T^2 \sigma))$.

5. Kink finding and incorporation into the lattice.

All sub-processes (sticking, diffusion, ES barrier crossing, nucleation, kink attachment) involve the same H-bond breaking/forming and therefore share the *same temperature dependence*. When the surface premelts at T_c, the QLL mobility increases, the ES barrier softens, nucleation becomes easier, and kink density rises—all simultaneously. This is why a single $\text{sech}^2(T_c)$ captures the envelope of all hopping dynamics. The Snow Crystal Equation is the *mean-field limit* of the full kinetic Monte Carlo on the ice lattice.

Acknowledgments

We thank the Libbrecht, Sazaki, and Murata research groups for the experimental data that made this analysis possible. Now as for Ken Libbrecht giving me a hard time when I was auditing physics classes at Caltech and haranguing him about "standing electromagnetic radiation" in snow crystals... my paper effectively obsoletes his SDAK equation. Hopefully he of all people can appreciate that in the snow crystal physics business, revenge is a dish best served cold!

References

[1] K. G. Libbrecht, "Physical Dynamics of Ice Crystal Growth," *Annu. Rev. Mater. Res.* **47**, 271–295 (2017). doi:10.1146/annurev-matsci-070616-124135.

[2] K. G. Libbrecht, "Toward a Comprehensive Model of Snow Crystal Growth Dynamics: 2. Structure Dependent Attachment Kinetics near -5 C," *arXiv:1302.1231* (2013).

[3] K. G. Libbrecht, "A Quantitative Physical Model of the Snow Crystal Morphology Diagram," *arXiv:1910.09067* (2019).

[4] K. G. Libbrecht, "The Physics of Snow Crystals," *Rep. Prog. Phys.* **68**, 855–895 (2005). doi:10.1088/0034-4885/68/4/R03.

[5] K. G. Libbrecht and R. Bell, "Chemical Influences on Ice Crystal Growth from Vapor," *arXiv:1101.0127* (2010).

[6] K. G. Libbrecht, "Explaining the Formation of Thin Ice Crystal Plates with Structure-Dependent Attachment Kinetics," *J. Cryst. Growth* **258**, 168–175 (2003). doi:10.1016/S0022-0248(03)01496-9.

[7] K. Murata, H. Asakawa, K. Nagashima, Y. Furukawa, and G. Sazaki, "Thermodynamic Origin of Surface Melting on Ice Crystals," *Proc. Natl. Acad. Sci. U.S.A.* **113**(44), E6741–E6748 (2016). doi:10.1073/pnas.1608888113.

[8] K. Murata, K. Nagashima, and G. Sazaki, "How Do Ice Crystals Grow inside Quasiliquid Layers?," *Phys. Rev. Lett.* **122**, 026102 (2019). doi:10.1103/PhysRevLett.122.026102.

[9] G. Miyamoto, A. Kouchi, K. Murata, K. Nagashima, and G. Sazaki, "Growth Kinetics of Elementary Spiral Steps on Ice Prism Faces Grown in Vapor and Their Temperature Dependence," *Cryst. Growth Des.* **22**(11), 6639–6646 (2022). doi:10.1021/acs.cgd.2c00851.

[10] M. Inomata, K. Murata, H. Asakawa, K. Nagashima, S. Nakatsubo, Y. Furukawa, and G. Sazaki, "Temperature Dependence of the Growth Kinetics of Elementary Spiral Steps on Ice Basal Faces Grown from Water Vapor," *Cryst. Growth Des.* **18**(2), 786–793 (2018). doi:10.1021/acs.cgd.7b01251.

[11] H. Fukazawa and S. Mae, "The Vibrational Spectra of Ice Ih and Polar Ice," in *Physics of Ice Core Records*, 25–42 (Hokkaido University Press, 2000). Handle: `https://hdl.handle.net/2115/32460`.

[12] J. G. Dash, H. Fu, and J. S. Wettlaufer, "The Premelting of Ice and Its Environmental Consequences," *Rep. Prog. Phys.* **58**(1), 115–167 (1995). doi:10.1088/0034-4885/58/1/003.

[13] U. Nakaya, *Snow Crystals, Natural and Artificial* (Harvard University Press, Cambridge, MA, 1954).

[14] P. H. Gammon, H. Kiefte, M. J. Clouter, and W. W. Denner, "Elastic Constants of Artificial and Natural Ice Samples by Brillouin Spectroscopy," *J. Glaciol.* **29**(103), 433–460 (1983). doi:10.3189/S0022143000030355.

[15] A. D. Fortes, "Accurate and Precise Lattice Parameters of H_2O and D_2O Ice Ih between 1.6 and 270 K from High-Resolution Time-of-Flight Neutron Powder Diffraction Data," *Acta Crystallogr. B* **74**(2), 196–216 (2018). doi:10.1107/S2052520618002159.

[16] P. Llombart, E. G. Noya, and L. G. MacDowell, "Surface Phase Transitions and Crystal Habits of Ice in the Atmosphere," *Sci. Adv.* **6**(21), eaay9322 (2020). doi:10.1126/sciadv.aay9322.

[17] M. A. Sánchez, T. Kling, T. Ishiyama, M.-J. van Zadel, P. J. Bisson, A. Mejía, M. J. Moreira de la Luz, F. J. Ruiz-Salvador, M. Mizukami, Y. Tsuchikawa, T. Kawasaki, S. Matsumura, B. J. Ramírez-Cuesta, and J. A. Padró, "Experimental and Theoretical Evidence for Bilayer-by-Bilayer Surface Melting of Crystalline Ice," *Proc. Natl. Acad. Sci. U.S.A.* **114**(2), 227–232 (2017). doi:10.1073/pnas.1612893114.

[18] M. Elbaum, S. G. Lipson, and J. G. Dash, "Optical Study of Surface Melting on Ice," *J. Cryst. Growth* **129**(3–4), 491–505 (1993). doi:10.1016/0022-0248(93)90483-D.

[19] A. Döppenschmidt and H.-J. Butt, "Measuring the Thickness of the Liquid-Like Layer on Ice Surfaces with Atomic Force Microscopy," *Langmuir* **16**(16), 6709–6714 (2000). doi:10.1021/la990799w.

[20] S. L. Bore and F. Paesani, "Realistic Phase Diagram of Water from 'First-Principles' Data-Driven Quantum Simulations," *Nat. Commun.* **14**, 3349 (2023). doi:10.1038/s41467-023-38855-1.

[21] A. K. Garg, "High-Pressure Raman Spectroscopic Study of the Ice Ih → Ice IX Phase Transition," *Phys. Status Solidi A* **110**(2), 467–480 (1988). doi:10.1002/pssa.2211100219.

[22] Ph. Pruzan, J.-C. Chervin, and M. Gauthier, "Raman Spectroscopy Investigation of Ice VII and Deuterated Ice VII to 40 GPa: Disorder in Ice VII," *Europhys. Lett.* **13**(1), 81–87 (1990). doi:10.1209/0295-5075/13/1/014.

[23] E. Yokoyama and T. Kuroda, "Pattern Formation in Growth of Snow Crystals Occurring in the Surface Kinetic Process and the Diffusion Process," *Phys. Rev. A* **41**(4), 2038–2049 (1990). doi:10.1103/PhysRevA.41.2038.

[24] W. W. Mullins and R. F. Sekerka, "Stability of a Planar Interface During Solidification of a Dilute Binary Alloy," *J. Appl. Phys.* **35**(2), 444–451 (1964). doi:10.1063/1.1713333.